



Lakshya JEE 2.0 (2024)

Matrices

DPP-01

1. Calculate the value of x, y, z & a which satisfied the following matrix equation

$$\begin{bmatrix} x+1 & 3y+x \\ z-1 & 4a-6 \end{bmatrix} = \begin{bmatrix} 0 & 7 \\ 3 & 2a \end{bmatrix}$$

2. Construct a 2×2 matrix whose elements are given

$$\text{by } a_{ij} = \frac{(i+2j)^2}{2}.$$

3. The matrix $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ is:

- (1) Column matrix (2) Null matrix
(3) Scalar matrix (4) Identity matrix

4. The matrix $\begin{bmatrix} -1 & 0 & 0 \\ 2 & -1 & 0 \\ 3 & -7 & -1 \end{bmatrix}$ is:

- (1) Upper triangular matrix
(2) Diagonal matrix
(3) Scalar matrix
(4) Lower triangular matrix

5. If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ a & b & -1 \end{bmatrix}$ then A^2 is equal to:

- (1) 0 (2) $-A$
(3) I (4) $2A$

6. If $1, \omega$ and ω^2 are the cube roots of unity and

$$\begin{bmatrix} 1+\omega & 2\omega \\ -2\omega & -b \end{bmatrix} + \begin{bmatrix} a & -\omega \\ 3\omega & 2 \end{bmatrix} = \begin{bmatrix} 0 & \omega \\ \omega & 1 \end{bmatrix}$$

then $a^2 + b^2$ is equal to:

- (1) $1 + \omega^2$ (2) $\omega^2 - 1$
(3) $1 + \omega$ (4) $(1 + \omega)^2$

7. If ω is a complex cube root of unity and

$$A = \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix} \text{ then } A^{50} \text{ is equal to:}$$

- (1) $\omega^2 A$ (2) ωA
(3) A (4) 0

8. If $x \begin{bmatrix} -3 \\ 4 \end{bmatrix} + y \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} 10 \\ -5 \end{bmatrix}$, then:

- (1) $x = -2, y = 1$ (2) $x = -9, y = 10$
(3) $x = 22, y = 1$ (4) $x = 2, y = -1$

9. If $\begin{bmatrix} 1 & x & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 5 & 1 \\ 0 & 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ 1 \\ -2 \end{bmatrix} = 0$, then the value of x is:

- (1) 0 (2) $2/3$
(3) $\frac{5}{4}$ (4) $-4/5$

10. If $A = \begin{bmatrix} 1 & -2 & -3 \\ -4 & 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 2 & 1 \end{bmatrix}$ then find the product AB and BA .

11. If $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, then $A^n, n \in N$ is

(1) $\begin{bmatrix} 3^n & 3^n & 3^n \\ 3^n & 3^n & 3^n \\ 3^n & 3^n & 3^n \end{bmatrix}$

(2) $\begin{bmatrix} 3n & 3n & 3n \\ 3n & 3n & 3n \\ 3n & 3n & 3n \end{bmatrix}$

(3) $\begin{bmatrix} 3^{n+1} & 3^{n+1} & 3^{n+1} \\ 3^{n+1} & 3^{n+1} & 3^{n+1} \\ 3^{n+1} & 3^{n+1} & 3^{n+1} \end{bmatrix}$

(4) $\begin{bmatrix} 3^{n-1} & 3^{n-1} & 3^{n-1} \\ 3^{n-1} & 3^{n-1} & 3^{n-1} \\ 3^{n-1} & 3^{n-1} & 3^{n-1} \end{bmatrix}$



12. Let a, b and c be three real numbers satisfying

$$[a \ b \ c] \begin{bmatrix} 1 & 9 & 7 \\ 8 & 2 & 7 \\ 7 & 3 & 7 \end{bmatrix} = [0 \ 0 \ 0] \dots\dots\dots (E).$$

Let ω be a solution of $x^3 - 1 = 0$ with $\text{Im}(\omega) > 0$. If $a = 2$ with b and c satisfying (E), then the value of

$$\frac{3}{\omega^a} + \frac{1}{\omega^b} + \frac{3}{\omega^c} \text{ is equal to:}$$

- (1) -2 (2) 2
(3) 3 (4) -3

13. Simplify $\tan \theta$

$$\begin{bmatrix} \sec \theta & \tan \theta \\ \tan \theta & -\sec \theta \end{bmatrix} + \sec \theta \begin{bmatrix} -\tan \theta & -\sec \theta \\ -\sec \theta & \tan \theta \end{bmatrix}$$

- (1) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (2) $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$
(3) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ (4) $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$

14. Which of following option is not correct.

- (1) Minimum number of zeroes in upper triangular matrix is equal to minimum number of zeroes in lower triangular matrix.
(2) Minimum number of zeroes in diagonal matrix is equal to $n(n-1)$
(3) Number of Zeroes in scalar matrix is equal to $n(n-1)$
(4) Number of zeroes in scalar matrix can be more than $n(n-1)$

15. Minimum number of zeroes in diagonal matrix A having 25 elements is m and diagonal matrix B having 16 elements is n then value of $m - n$ is :

- (1) 8 (2) 12
(3) 15 (4) 20

16. Let $A = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}, (a \in \mathbb{R})$ such that

$$A^{32} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \text{ Then a value of } \alpha \text{ is.}$$

- (1) $\pi/16$ (2) 0
(3) $\pi/32$ (4) $\pi/64$

17. If A is a square matrix such that $A^2 = A$ and $B = I - A$, then $AB + BA + I - (I - A)^2 =$

- (1) A (2) $2A$
(3) $-A$ (4) $I - A$

18. If A and B are two matrices such that $AB = A$ and $BA = B$, then B^2 is equal to

- (1) B (2) A
(3) 1 (4) 0



Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. $x = -1; y = 8/3, z = 4, a = 3$

2. $\begin{bmatrix} \frac{9}{2} & \frac{25}{2} \\ \frac{2}{8} & \frac{2}{18} \end{bmatrix}$

3. (3)

4. (4)

5. (3)

6. (3)

7. (2)

8. (1)

9. (3)

10. $AB = \begin{bmatrix} -12 & -10 \\ 10 & 3 \end{bmatrix}, BA = \begin{bmatrix} -10 & 2 & 9 \\ -16 & 2 & 13 \\ -2 & -2 & -1 \end{bmatrix}$

11. (4)

12. (1)

13. (4)

14. (4)

15. (1)

16. (4)

17. (1)

18. (1)



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